

# Vydhra AI Agents Engineering Bootcamp

Duration: 8 Weeks

Format: Agent-First  
Design

Level: Intermediate to  
Production

## Advanced AI Agents Engineering Roadmap

|                |                                                                         |   |
|----------------|-------------------------------------------------------------------------|---|
| Module 01      | Week 1 - Foundations of AI Agents                                       | ■ |
| Topics Covered | AI agents vs LLM apps, real-world use cases, and agent architecture     |   |
|                | Reason-act-observe loops, tools, APIs, and system design essentials     |   |
| Practice       | Build a basic AI assistant using an LLM API                             |   |
| Outcome        | Students understand agent architecture and build a first working agent. |   |

|                |                                                                       |   |
|----------------|-----------------------------------------------------------------------|---|
| Module 02      | Week 2 - Prompt Engineering and Control                               | ■ |
| Topics Covered | Chain-of-thought, ReAct, few-shot prompting, and system prompt design |   |
|                | Tool calling, structured outputs, prompt debugging, and optimization  |   |
| Practice       | Build a reasoning-based agent with tool usage                         |   |
| Outcome        | Students control LLM behavior more reliably and intelligently.        |   |

|                |                                                                     |   |
|----------------|---------------------------------------------------------------------|---|
| Module 03      | Week 3 - LangChain and Tool-Based Agents                            | ■ |
| Topics Covered | Chains, tools, memory, execution flow, and decision-making patterns |   |
|                | Multi-tool agents with search, calculator, and API integrations     |   |
| Practice       | Build a multi-tool agent using LangChain                            |   |
| Outcome        | Students build modular agents using tools and memory.               |   |

|                |                                                                   |   |
|----------------|-------------------------------------------------------------------|---|
| Module 04      | Week 4 - RAG Systems                                              | ■ |
| Topics Covered | Embeddings, vector databases, chunking strategies, and retrieval  |   |
|                | Retrieval optimization and hallucination-reduction techniques     |   |
| Practice       | Build a PDF knowledge chatbot using RAG                           |   |
| Outcome        | Students create knowledge-based AI systems for real applications. |   |

| Module 05      | Week 5 - LangGraph Workflows                                        | ■ |
|----------------|---------------------------------------------------------------------|---|
| Topics Covered | State machines, planner-executor flows, and looping agent workflows |   |
|                | Fallbacks, error handling, and structured multi-step design         |   |
| Practice       | Build a planner-executor agent system                               |   |
| Outcome        | Students design reliable multi-step AI workflows.                   |   |

| Module 06      | Week 6 - Multi-Agent Systems                                     | ■ |
|----------------|------------------------------------------------------------------|---|
| Topics Covered | AutoGen, CrewAI, collaboration patterns, and agent communication |   |
|                | Task delegation, role-based agents, and system scaling concepts  |   |
| Practice       | Build a research-plus-writer multi-agent system                  |   |
| Outcome        | Students build collaborative AI systems with multiple agents.    |   |

| Module 07      | Week 7 - Production AI Systems                                  | ■ |
|----------------|-----------------------------------------------------------------|---|
| Topics Covered | Deployment with FastAPI, logging, monitoring, and observability |   |
|                | Cost optimization, retries, guardrails, and safety handling     |   |
| Practice       | Deploy an AI agent web app with logging and monitoring          |   |
| Outcome        | Students build production-ready AI applications.                |   |

| Module 08      | Week 8 - Evaluation and Capstone                                 | ■ |
|----------------|------------------------------------------------------------------|---|
| Topics Covered | Evaluation, debugging, tracing, benchmarks, and optimization     |   |
|                | Final capstone: full AI agent system with deployment and testing |   |
| Practice       | Build a final portfolio project with rigorous evaluation         |   |
| Outcome        | Students complete a portfolio-ready, production-grade AI system. |   |